Interactive A* Pathfinding Puzzle Game

Demonstrate how different heuristics affect A* search efficiency while preserving path optimality in grids with varying terrain costs

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ABSTRACT

An interactive browser-based tool running Dijkstra, A∗ Manhattan, and A∗ Euclidean side-by-side on weighted grids. Users paint walls (∞), grass (×2), and swamps (×5), then watch all three execute with per-cell f(n) cost overlays. Benchmarks confirm A* cuts node expansions by ~24% vs Dijkstra on identical paths.

BACKGROUND & MOTIVATION

Algorithm trade-offs are hard to internalize from pseudocode alone — the contrast between uninformed and informed search only becomes intuitive when you can watch both exploration frontiers side by side in real time.

Most tools show one algorithm at a time on uniform-cost grids, making direct comparison impossible. This project closes that gap: Dijkstra, A∗ Manhattan, and A∗ Euclidean run simultaneously on shared weighted grids, with step-by-step replay and per-cell f/g/h overlays that reveal exactly why each algorithm expands each node.

Student feedback during development confirmed the core gap: watching A∗'s frontier stay focused while Dijkstra fans out in all directions builds intuition no static textbook diagram can replicate.

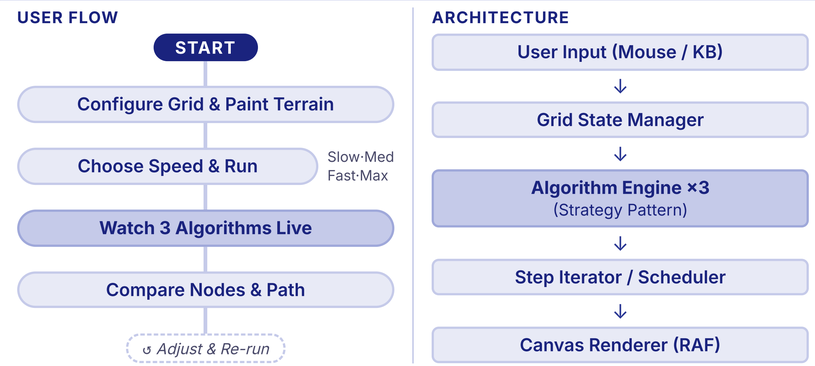
ALGORITHM OVERVIEW

Algorithm	Heuristic h(n)	Best suited for
Dijkstra	`0 (uninformed)`	Baseline; always optimal
A∗ Manhattan	`|Δx| + |Δy|`	Square 4-way grids
A∗ Euclidean	`√(Δx² + Δy²)`	Any grid geometry

CONCLUSION & FUTURE WORK

The visualizer confirms heuristic choice has a measurable impact on efficiency. Learners build intuition through direct manipulation, not passive reading. Planned: hex grid support, bidirectional A∗, CSV export.

TOOL USER FLOW & SYSTEM ARCHITECTURE



INTERACTIVE FEATURES

- **Terrain Painting** — Drag to paint Walls (∞), Grass (×2), or Swamp (×5) across all three panels; right-click to erase
- **Draggable Endpoints** — Reposition Start / Goal at any time; re-runs automatically on next playback
- **Maze & Preset Generator** — Hand-crafted Maze, random scatter, or barrier presets — one toolbar click
- **Playback Controls** — Run / Pause / Step; four speeds from Slow (300 ms) to Max (full-batch, no delay)
- **f(n) Overlay** — Toggle per-cell f(n) cost overlays over each panel for deep algorithmic inspection
- **Live Metrics** — Nodes-expanded counter and path length updated live
- **Layer 1 — The Trap** — Max 5 walls; craft barriers to lure A∗ (Manhattan) into a local-optimum dead end
- **Layer 2 — The Swamp** — Pink brush predicts your path; system compares Predict Cost vs A∗ Path Cost and gives a verdict

VISUALIZATION DESIGN & COLOR SYSTEM

Purpose-built palette encodes cell states at a glance:

Start (S) · Goal (G) · Open set · Expanded · Optimal path · Grass (×2) · Swamp (×5) · Wall (∞)

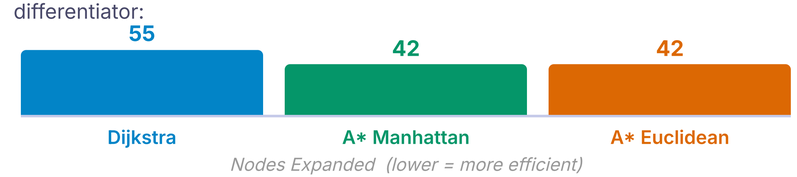
Algorithm panels share Start, Goal and Path colors so results are directly comparable. Open-set blue reveals the live search frontier; expanded orange marks already-processed nodes. Terrain colors are visually distinct from algorithm-state colors to prevent misreading. A dynamic sine-wave neon pulsing effect highlights optimal paths for visual clarity.

TOOL DEMO — THREE ALGORITHMS RUNNING IN PARALLEL

Dijkstra (left) expands more nodes than A∗ Manhattan (centre) and A∗ Euclidean (right); all three find the same optimal path (violet).

PERFORMANCE RESULTS & BENCHMARK

Benchmark on a 20×20 square grid (seed = 42). All three algorithms return the **same optimal path of length 37**. Node expansion is the key efficiency differentiator:



Algorithm	Path Length	Nodes Expanded
Dijkstra	37	55
A∗ Manhattan	37	**42** ↓ 24%
A∗ Euclidean	37	**42** ↓ 24%

KEY INSIGHT

A* reduces node expansions by ~24% compared to Dijkstra while always guaranteeing the same optimal path — a direct, measurable benefit of informed search with an admissible heuristic.